

ProjectDesignPhase-II

SolutionRequirements(Functional&Non-functional)

Date	7 march 2026
TeamID	SWTID-2026-7607
ProjectName	MicroGoal
MaximumMarks	4Marks

FunctionalRequirements:

Followingarethefunctionalrequirementsoftheproposedsolution.

FR No.	FunctionalRequirement(Epic)	SubRequirement(Story/Sub-Task)
FR-1	UserAuthentication&AccountManagement	UserRegistration
		UserLogin
		PasswordEncryption(Hashing)
		JWTTokenGeneration
FR-2	GoalManagement(CRUDOperations)	CreateGoal
		ViewAllGoals(User-Specific)
		UpdateGoal
		DeleteGoal
FR-3	Authorization&AccessControl	ProtectRoutesusingAuthMiddleware

		RestrictUsersfromAccessingOthers'Data
FR-4	ErrorHandling&ResponseManagement	CentralizedErrorMiddleware
		ProperHTTPStatusCodes(200,201,400,401,404,500)
FR-5	APITesting&Validation	TestEndpointsusingThunderClient
		ValidateRequestBodyData

Non-functionalRequirements:

Followingarethennon-functionalrequirementsoftheproposedsolution.

NFR No.	Non-Functional Requirement	Description
NFR-1	Usability	APIendpointsshouldbesimple,clear,andeasytounderstandfordevelopers.
NFR-2	Security	AllAPIrequestsmustbesecuredusingJWTauthentication. Passwordsmust be hashed before storing.
NFR-3	Reliability	Thesystemshouldhandleerrorsproperlyandavoidcrashesduringinvalid requests.
NFR-4	Performance	APIresponsesshouldbereturnedwithin2secondsundernormal load conditions.

NFR-5	Availability	The backend servers should be accessible 24/7 with minimal downtime.
NFR-6	Scalability	The system should support increasing users and goals without performance degradation. The architecture should allow future feature expansion.